Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I think that i will use a concept similar to the one used in the “pet” project where there were multiple parts that determined the pet’s mood, but some of them were ranked more heavily in determining the mood, i can use this idea to rank the important of each category when deciding what game best fits the user’s preferences. I will be using dataclasses to store information about each game, and lists to store the games and their characteristics, as well as loops to search through the lists. Conditionals can be used to determine which game best matches the users preferences. And functions and drafter will be used to create the website itself. I want to get better at organizing websites with drafter and learn how to change some of the visual aspects. I also want to get better at modifying states across multiple routes

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part will be designing the layout of the website, and creating the lists of games and their characteristics. I think the hardest part will probably be creating the program to rank preference strength, and actually return an answer

based on the user's input, especially keeping track of the data across routes

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think creating the ranking system/recommendation system will take the most time and definitely the most effort. The dataclasses and game database will probably be implemented first

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

My phone and friends may be distractions for working on this project, but i will minimize them by setting specific time aside to work on the project, and having other time to do fun things, that way i can focus and still know that later i will have time for other activities. I will most likely need access to my laptop, the blockply textbook/my notes, and possibly the TA's/professors and possible LLMs. at this time i do not have any questions about the assignment requirements

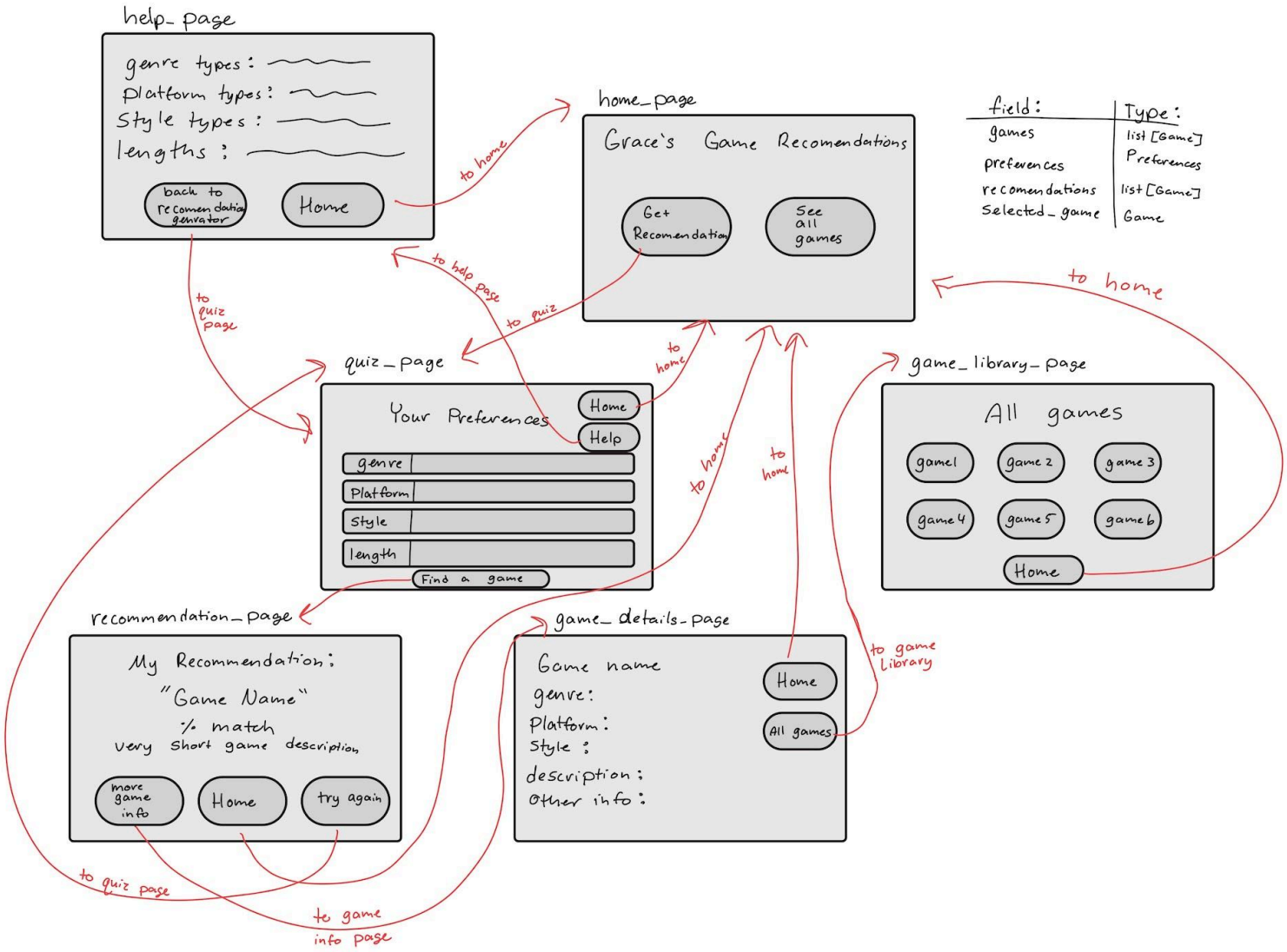
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


help_page
genre types:
platform types:
Style types:
lengths:
back to recomendation genrator
Home
to home
home_page
Grace's Game Recomendations
Get Recomendation
See all games
field:
Type:
games
list [Game]
preferences
Preferences
recomendations
list [Game]
Selected_game
Game
to help page
to quiz
to quiz page
to home
quiz_page
Your Preferences
Home
Help
genre
Platform
Style
length
Find a game
to home
game_library_page
All games
game1
game 2
game 3
game 4
game 5
game 6
Home
to home
to home
recommendation_page
My Recommendation:
"Game Name"
% match
very short game description
more game info
Home
try again
game_details_page
Game name
genre:
Platform:
Style:
description:
other info:
Home
All games
to game library
to quiz page
to game info page

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**A website to help people find game recommendations based on their personal preferences from a library of my favorite games**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	games: list[Game] preferences: Preferences recommendations: list[Game] selected_game: Game
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Dataclass: Preferences genre: str platform: str difficulty: int multiplayer: bool
Routes (minimum 7) The names of your route functions	home_page quiz_page submit_quiz recommendations_page game_library_page

	filter_games game_details_page back_to_home help_page
Pages (minimum 5) A brief description of the pages in your site	Home page: name and description of website, buttons to go start the quiz or see the full list of games Quiz page: where users input their preferences (genre, platform, difficulty, and style), button to submit preferences, and button to help page with accepted answers Help page: shows accepted answers for preferences page, button to return home or return back to quiz Recommendations page: shows the recommended game, the % match, and a short description, button to see more game information, button to return home, button to try quiz again Game library page: shows all games, each as a button to see more information about the game Game details page: displays detailed information about each game, including genre, difficulty, style, length, etc, and any personal notes from the developer, button to return home
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Genre: textbox Platform: selectbox difficulty : select box Style: checkbox (multiplayer game)
If Statements (minimum 3) Brief description of what will be evaluated in your if statements	If a game's genre matches the users preference, it adds a point to the recommendation score If a game's platform matches the users preference, it

and how they will meaningfully change the logic of the site	adds a point to the recommendation score If a game's difficulty matches the users preference, it adds a point to the recommendation score If a game's multiplayer style matches the users preference, it adds a point to the recommendation score If the users input is not an available option, and error message is returned and the user is promoted to attempt a different answer
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	A for loop will be used to iterate through the games list in the state, and it will be comparing the games characteristics to the users given preferences and uses conditional statements to create a match score for each game and the game with the greatest match score is returned as the recommended game

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I did not use any LLM's for this portion of the project

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.